Judy Sihyon Park

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Education

Carnegie Mellon University 2024 – present
Ph.D. in Engineering and Public Policy Pittsburgh, PA, United States
Advisors: Dr. Valerie Karplus & Dr. Nick Muller

McGill University 2016 – 2021
B.Eng. in Honours Mechanical Engineering Montreal, QC, Canada
Graduated with Distinction
Undergraduate Thesis Title: Combustion Instabilities of Flames in Metal Particle Suspensions (advised by Dr. Jeffrey Bergthorson)

References

Dr. Valerie Karplus (co-advisor)
Professor of Economics, Engineering, and Public Policy
Carnegie Mellon University
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(617) 800-7982

Dr. Nick Muller (advisor)
Professor of Economics, Engineering, and Public Policy
Carnegie Mellon University
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Peer-Reviewed Publications

Park, J. S, Muller, N. Z, and Karplus, V. J. (2026) *Technological Change in Steel Production: An Integrated Assessment of Climate Change, Air Quality, and Distributional Impacts*. (Work in Progress)

Palecka, J, **Park, J. S**, Goroshin, S, and Bergthorson, J. M. (2021) *Aluminum-propane-air hybrid flames in a Hele-Shaw cell. Proceedings of the Combustion Institute 38: 4461–4468*. DOI: 10.1016/j.proci.2020.09.017

Academic/Conference Presentations *(*presenting author)*

Park, J. S*, Muller, N. Z, Karplus, V. J. (May 2026) *Technological Change in Iron and Steel Production*. Association of Iron and Steel Technology Conference (AISTech), Pittsburgh, PA (Oral Presentation)

Park, J. S*, Muller, N. Z, Karplus, V. J. (Oct 2025) *Technological Change in Iron and Steel Production: Implications for Greenhouse Gas Emissions, Air Pollution, and Community Health*. INFORMS, Atlanta, GA (Invited Oral Presentation)

Park, J. S*, Muller, N. Z, Karplus, V. J. (May 2025) *Mortality and Distributional Effects of Air Pollution from Integrated Steelmaking in the United States*. Association of Iron and Steel Technology Conference (AISTech), Nashville, TN (Poster Presentation)

Park, J. S*, Muller, N. Z, Karplus, V. J. (Mar 2025) *Mortality and Distributional Effects of Air Pollution from Integrated Steelmaking in the United States*. Carnegie Mellon University Energy Week, Pittsburgh, PA (Poster Presentation – *2nd Place Winner*)

Park, J. S*, Palecka, J, Goroshin, S, and Bergthorson, J. M. (Aug 2021) *Combustion Instabilities of Flames in Metal Particle Suspensions*. Summer Undergraduate Research in Engineering, Montreal, QC (Poster Presentation)

Palecka, J*, **Park, J. S,** and Bergthorson, J. M. (Aug 2021) *Aluminimum-Propane-Air Flames in a Hele-Shaw Cell*. Adelaide, Australia (Oral Presentation)

Park, J. S*, Palecka, J, and Bergthorson, J. M. (Aug 2020) *Improving an Experimental Apparatus for the Study of Flame Instabilities in Zero-Carbon Metal Fuel Suspensions*. Summer Undergraduate Research in Engineering, Montreal, QC (Poster Presentation)

Invited Talks on Academic Work

2026 Center for Atmospheric Particle Studies Seminar, Carnegie Mellon University
2025 Direct-Reduced Iron Technology Committee Meeting, Association of Iron and Steel Technology

Awards & Fellowships

2025	Presidential Fellowship in the College of Engineering, Carnegie Mellon University	$87,700
2024	Carnegie Mellon University Energy Week Poster Contest – 2nd Place Winner	$750
2023	Massachussets Institute of Technology Policy Hackathon – 1st Place Winner	$125
2020	Mitacs/DAAD RISE-Globalink Research Award (for summer research at TU Berlin, cancelled due to COVID)	$6,000
	NSERC Undergraduate Student Research Award (for summer research at McGill University)	$8,900
2019	NSERC Undergraduate Student Research Award (for summer research at McGill University)	$8,900
	Outstanding Teaching Assistant Award, McGill University Department of Engineering	$500
	ELATE Scholar Award, McGill University	$1,000
2017	Tomlinson Engagement Award for Mentoring, McGill University	$300
2016	J.W. McConnell Entrance Scholarship, McGill University	$3,000
	British Columbia Achievement Scholarship	$1,250

Teaching Experience

Undergraduate Teaching Assistant, McGill University — 2017 – 2019
- Math 264: Advanced Calculus for Engineers — Fall 2018, Fall 2019
- Math 263: Ordinary Differential Equations for Engineers — Spring 2018
- Chem 120: General Chemistry 2 — Spring 2017

Tutor, McGill Engineering Peer Tutoring Service *(awarded Student Service of the Year in 2019)* — 2018 – 2020
- Math 264: Advanced Calculus for Engineers — Sep 2018 – Apr 2020
- Math 263: Ordinary Differential Equations for Engineers — Sep 2018 – Apr 2020
- Math 262: Intermediate Calculus for Engineers — Sep 2018 – Apr 2020

Professional Experience

Graduate Research Assistant, Carnegie Mellon University — 2024 – present
for Dr. Valerie Karplus and Dr. Nick Muller — Pittsburgh, PA
Researching climate change and air pollution impacts of hard-to-decarbonize industries

Mechanical Engineering Specialist, AtkinsRealis — 6/2023 – 6/2024
Mining & Metals North America — Toronto, ON
Mechanical/process design and project management for industrial air pollution control equipment

Mechanical Engineer-in-Training, Hatch Ltd. — 8/2021 – 6/2023
Technologies Group (previously Furnace Group) — Toronto, ON
Mechanical/process design and project management for metal smelting furnaces and related equipment

Undergraduate Research Assistant, McGill University — 5/2019 – 7/2021
for Dr. Jeffrey Bergthorson (Alternative Fuels Laboratory) — Montreal, QC
Researched combustion instabilities in metal micro-particles to examine their potential as e-fuels

Undergraduate Research Assistant, McGill University — 6/2018 – 8/2018
for Dr. Sabrina Leslie (Leslie Lab) — Montreal, QC
Conducted literature reviews to assist the development of a single-molecule microimaging device

Service & Mentorship

Association of Iron and Steel Technology (AIST)
- Association of Iron and Steel Technology Conference (AISTech), *Session Chair* — 2026
- Direct-Reduced Iron Technology Committee, *Young Professionals Chair* — 2025 – 2026

Carnegie Mellon University
- Student Facilitation Board, Dept. of Engineering and Public Policy, *Diversity, Equity, and Inclusion Chair* — 2025
- GRASP Admissions Mentoring Program, *Mentor* — 2024 – present

Hatch Ltd.

Diversity, Equity, and Inclusion Committee, *Member* 2021 – 2023

McGill University

Engineering Undergraduate Society, *Year Representative* 2016 – 2019

McGill Association of Mechanical Engineers, *Year Representative & Social Committee Chair* 2017 – 2019

Promoting Opportunities for Women in Engineering, *Member & Conference Volunteer* 2017 – 2018

McGill Association of Mechanical Engineers, *Year Representative & First Year Committee Chair* 2016 – 2017

Faculty of Engineering, *Orientation Week Leader* 2016 – 2017

Professional Affiliations and Certifications

Association of Iron and Steel Technology

Professional Engineers Ontario (EIT Certified & requirements met for P. Eng.)

American Society for Metals

The Minerals, Metals & Materials Society

The American Ceramic Society

Additional Information

Programs and Languages: Python, Matlab, R, ANSYS, LaTeX

Languages: English (native/bilingual), Korean (native/bilingual), Mandarin Chinese (HSK Level 3), French (DELF A1)